

1. Administrative & Study Identification

Full Study Title: Prevalence of Emerging Treatment-induced Mutations in Metastatic ER-positive Breast Cancer **Short Title/ Acronym:** Pangeia-2 **Protocol Number:** D3612L00003 **NCT Number:** NCT06417801 **Sponsor Name:** AstraZeneca **Investigational Product:** N/A (Observational biomarker study using QIAseq Targeted DNA Custom Panel) **Phase of Development:** N/A **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the prevalence of emerging *ESR1* mutations via liquid biopsy in two cohorts of patients with metastatic HR+/HER2- breast cancer (with and without prior therapies in the metastatic setting) in comparison with the patient's baseline *ESR1* mutation status as defined by primary tissue profiling. **Secondary Objectives:** To determine the prevalence of emerging treatment-induced mutations in other key genes including *PIK3CA*, *AKT1*, *PTEN*, *BRCA1*, *BRCA2*, *PALB2*, and *ERBB2*.

2.2 Background & Rationale To address the knowledge gap regarding the emergence of resistance mutations (such as *ESR1*) during the course of standard therapies in metastatic HR+/HER2- breast cancer. Understanding the genomic evolution and treatment-induced mutational shifts via non-invasive liquid biopsy is crucial for identifying resistance mechanisms, assessing the efficacy of CDK inhibitors and hormone therapy combinations, and guiding subsequent personalized treatment decisions.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational **Control Type:** N/A (Cohort study) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 70 participants **Number of Sites:** Global and Regional AstraZeneca partner sites **Estimated Study Start:** 17-Jun-2024 **Estimated Primary Completion:** 30-Dec-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 to 100 years. 2. Confirmed diagnosis of locally-advanced or metastatic HR+/HER2- breast

cancer. 3. For Cohort 1: Treatment naive or previously exposed to adjuvant therapies, with no prior palliative therapy, and candidates to receive first-line hormone therapy. 4. For Cohort 2: Progression during hormone therapy plus CDK inhibitor treatment. 5. Primary tumor tissue must be available for testing.

4.2 Exclusion Criteria 1. Patients who are NOT able to provide informed consent. 2. Lack of available primary tumor tissue for baseline NGS profiling.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: N/A

(Observational). Patients undergo genomic profiling via DNA extraction from FFPE tissue and plasma utilizing the QIASymphony extraction kits. Next-Generation Sequencing (NGS) libraries are prepared using the QIAseq Targeted DNA Custom Panel and sequenced on Illumina platforms (NextSeq for liquid biopsy, MiSeq for tissue). **Route of Administration:** N/A (Standard of care hormone and CDK inhibitor therapies as prescribed; blood draws for liquid biopsy). **Duration of Treatment:** Observational follow-up extending until disease progression or study completion.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care systemic and localized therapies for metastatic HR+/HER2- breast cancer (e.g., hormone therapies, CDK inhibitors). **Prohibited:** N/A for observational setting.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, Confirmation of HR+/HER2- status
Baseline	Day 0	FFPE primary tumor tissue retrieval, baseline liquid biopsy (plasma) collection
Interim	Aligned with standard of care	Follow-up liquid biopsy collection (e.g., upon disease progression), ongoing monitoring of clinical response
End of Study	Dec 2027	Final genomic data extraction, clinical outcome correlations, and study closure

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Prevalence of emerging *ESR1* mutations detected in liquid biopsy (plasma) compared to the baseline *ESR1* mutation status derived from primary tumor tissue profiling. **Measurement Method:** Next-Generation Sequencing (NGS) GS Focus Liquid Breast and GS Focus Breast assays.

7.2 Secondary & Exploratory Endpoints Prevalence and correlation of emergent mutations in a custom panel of genes including

PIK3CA, AKT1, PTEN, ERBB2, BRCA1, BRCA2, and PALB2 using matched liquid and tissue biopsies.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine clinical monitoring for AEs associated with standard of care therapies and blood draw procedures. **Serious Adverse Events (SAEs):** Assessed according to standard institutional clinical practice guidelines. **Data Monitoring Committee (DMC):** Managed internally by the Sponsor's observational research data teams.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Enrolled patients with valid NGS results from both baseline tissue and matched liquid biopsy samples. **Sample Size Justification:** $N = 70$ calculated to provide adequate precision for estimating the prevalence of emerging *ESR1* mutational shifts in the selected populations. **Handling of Missing Data:** Patients with insufficient DNA yield or NGS failures will be excluded from the matched-pair mutational analysis.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Validated central laboratory biomarker assessment. The NGS panels were internally validated for 100% sensitivity, specificity, and accuracy. **Audit Trail:** Monitored strict limits of detection (LOD): 0.5% Variant Allele Frequency (VAF) for the liquid biopsy panel and 5% VAF for the tissue panel.

11. Study Identifiers & Governance

Primary ID: {{D3612L00003}} **Registry Number:** {{NCT06417801}}
Sponsor/Lead Organization: {{AstraZeneca}} **Study Title:** {{Pangeia-2}} **Official Regulatory Title:** {{Prevalence of Emerging Treatment-induced Mutations in Metastatic ER-positive Breast Cancer}}

12. Clinical Framework (Metastatic Breast Cancer Specific)

Primary Condition: {{Locally-advanced/metastatic HR+/HER2- breast cancer}} **Diagnosis Threshold:** {{Histologically confirmed ER-

positive breast cancer with available baseline tumor tissue}}
Medical Methodology: {{Matched tissue and liquid biopsy (plasma)
Next-Generation Sequencing (NGS)}} **Optimization Target:**
{{Detection of emerging ESR1 and broader custom panel resistance
mutations}}

13. Study Procedures & Methodology

Management Pathway: {{Standard clinical pathway for HR+/HER2-
metastatic breast cancer (hormone therapy +/- CDK inhibitors)}}
Education Protocol (HCP): {{Training on FFPE block retrieval and
plasma processing using QIASymphony extraction}} **Education
Protocol (Patient):** {{Informed consent covering genomic profiling
and liquid biopsy serial blood draws}} **Quality Audit Mechanism:**
{{NGS performance validation tracking (100% accuracy validation;
0.5% LOD VAF liquid / 5.0% LOD VAF tissue)}}

14. Observation & Follow-Up Schedule

Total Duration: {{Follow-up extending to December 2027}} **Onsite
Visit Cadence:** {{Synchronized with standard of care oncology
visits}} **Remote Monitoring Frequency:** {{N/A for direct
observational clinical data extraction}} **Checklist Review
Interval:** {{Ongoing extraction at progression milestones}}

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				